7.3.3 Applications

Example 7.26

A truck travels on a toll road with a speed limit of 80 km/hr. The truck completes a 164 km journey in 2 hours. At the end of the toll road the trucker is issued with a speed violation notice. Justify this using the Mean Value Theorem.

Solution

Let $f(t)$ be the distance travelled by the trucker in 't' hours. This is a continuous function in $[0,2]$ and differentiable in $(0,2)$. Now, $f(0)=0$ and $f(2)=164$. By an application of the Mean Value Theorem, there exists a time c such that,

$$f'(c)=\frac{164-0}{2-0}=82>80.$$

Therefore at some point of time, during the travel in 2 hours the trucker must have travelled with a speed more than 80 km/hr which justifies the issuance of a speed violation notice. ■

Example 7.27

Suppose $f(x)$ is a differentiable function for all x with $f'(x)\le 29$ and $f(2)=17$. What is the maximum value of $f(7)$?

Solution

By the mean value theorem we have, there exists $'c'\in(2,7)$ such that,

$$\frac{f(7)-f(2)}{7-2} = f'(c)\le 29.$$

$$\text{Hence, } f(7)\le 5\times 29+17 = 162$$

Therefore, the maximum value of $f(7)$ is 162. ■

Example 7.28

Prove, using mean value theorem, that

$$|\sin\alpha-\sin\beta|\le|\alpha-\beta|,\ \alpha,\ \beta\in\mathbb{R}.$$

Solution

Let $f(x)=\sin x$ which is a differentiable function in any open interval. Consider an interval $[\alpha,\beta]$. Applying the mean value theorem there exists $c\in(\alpha,\beta)$ such that,

$$\frac{\sin\beta-\sin\alpha}{\beta-\alpha} = f'(c)=\cos(c)$$

$$\text{Therefore, } \left|\frac{\sin\alpha-\sin\beta}{\alpha-\beta}\right| = |\cos(c)|\le 1$$

$$\text{Hence, } |\sin\alpha-\sin\beta|\le|\alpha-\beta|.$$ ■

Remark

If we take $\beta=0$ in the above problem, we get $|\sin\alpha|\le|\alpha|$.

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